

Montrac System Description

Manual

Lists of available machines, tools, bases and coordinates.

W1/W2 Montrac

V2

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SYSTEM OVERVIEW

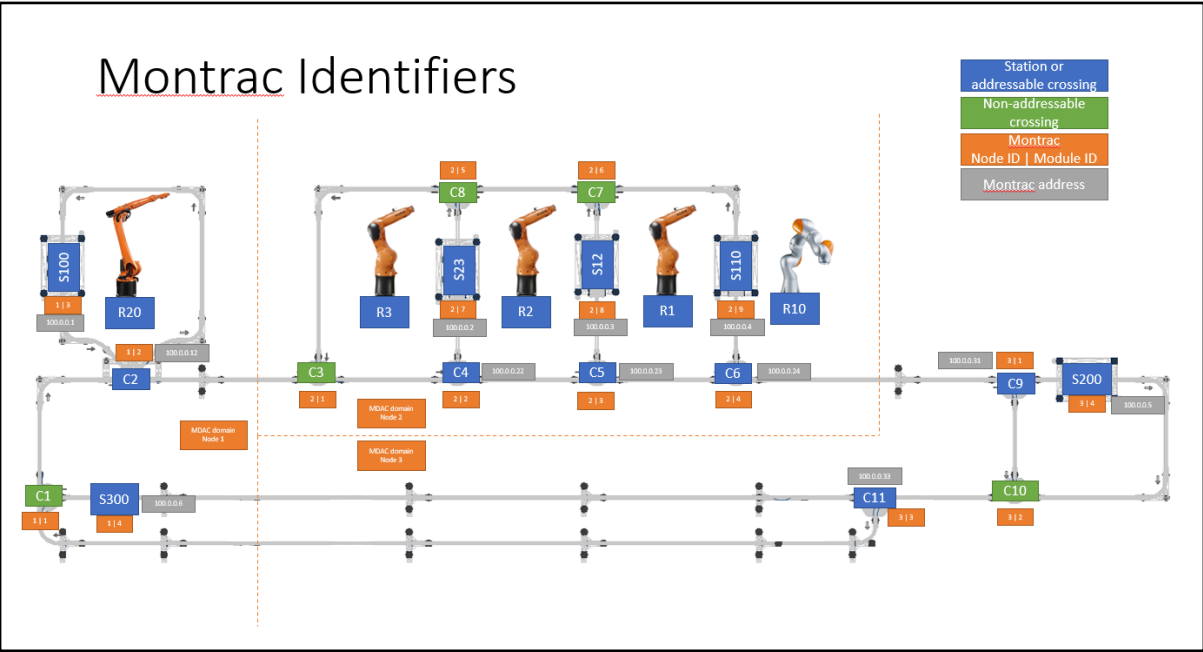


Figure 1: Montrac system map

ROBOTS

There are 4 robots on the monorail. All robots support all operations.

Numerical ID	String ID
1	R1
2	R2
3	R3
20	R20

Table 1: List of available robots.

All robots have different bases, to ensure the same set of positions can be used for different robots, to perform the same task. Base IDs are used as inputs to the robot operations, depending on where the operation should be performed and on which robot.

BASES - R1

Base ID	Location
1	Station S110
2	Station S12
5	Table corner next to station S110
6	Table corner next to station S12

Table 2: List of available bases on robot R1.

BASES - R2

Base ID	Location
1	Station S12
2	Station S23
5	Table corner next to station S12
6	Table corner next to station S23

Table 3: List of available bases on robot R2.

BASES - R3

Base ID	Location
1	Station S23
5	Table corner next to station S23
6	Table corner next to the AGV docking station

Table 4: List of available bases on robot R3.

BASES - R20

Base ID	Location
1	AGV docking station
2	Station S100

Table 5: List of available bases on robot R20.

STATIONS

There are 6 stations available on the monorail. Shuttles can be sent between them in any way.

Station Name	Usage	Location
S100	Robot Operations	Next to R20 robot
S200	Manual Operations / Loading / Unloading	At the back of the monorail
S300	Parking station / Shuttle Storage	Next to the windows
S110	Robot Operations	Next to R1 robot
S12	Robot Operations	Between R1 and R2 robots
S23	Robot Operations	Between R2 and R3 robots

Table 6: List of available stations.

SHUTTLES

There are 6 shuttles with interchangeable pallets. There are no differences between the shuttles, other than their respective ID.

Shuttle ID
1
2
3
4
5
6

Table 7: List of available shuttles.

Pallet ID	Function
1	Transports RC Car crate
2	Transports RC Car crate
3	Transports RC Car chassis
4	Transports RC Car chassis
5	Empty
6	Holds calibration L piece + point

Table 8: List of available shuttle pallets.

Pallets and shuttles with the same ID are usually paired together, but that is not needed. Always check that shuttles and pallets are paired as required before running any automated scripts.

TOOLS

Electric tools can be used with any robot. Pneumatic tools only either work with robots R1, R2 and R3, or just with robot R20, due to pneumatic connections.

Tool ID	Name	Type
1	Calibration Needle	N/A
2	EGI 40	Electric
3	EGL 90	Electric
7	EGL 90	Electric
8	Large Pneu	Pneumatic
9	Large Pneu	Pneumatic
12	Small Pneu	Pneumatic
16	EGI 80	Electric

Table 9: List of available robot tools.

RC CAR POSITIONS

For RC car assembly and disassembly, robot positions were defined in a JSON structure. These values are valid for all robots and appropriate bases. X Y Z values are in millimeters. Rotation values R are in degrees.

```
{
  "BatteryPCB": {
    "ShuttlePallet": {
      "X": -43.19,
      "Y": 69.00,
      "Z": 37.03,
      "R": -90.0
    },
    "ShuttleCar": {
      "X": 65.0,
      "Y": 49.89,
      "Z": 47.0,
      "R": 0.0
    }
  },
  "MotorPCB": {
    "ShuttlePallet": {
      "X": -43.19,
      "Y": 69.00,
      "Z": 10.30,
      "R": -90.0
    },
    "ShuttleCar": {
      "X": 98.34,
      "Y": 51.44,
      "Z": 58.03,
      "R": 0.0
    }
  },
  "Pallet": {
    "Shuttle": {
      "X": -43.19,
      "Y": 65.43,
      "Z": 19.08,
      "R": 0.0
    },
    "TablePos1": {
      "X": 44.63,
```



```
        "Y": 100.20,  
        "Z": 36.90,  
        "R": 0.0  
    },  
    "TablePos2": {  
        "X": 154.53,  
        "Y": 100.19,  
        "Z": 37.04,  
        "R": 0.0  
    }  
}  
}
```